



Research Engineer Technical Screening Challenge

School of Mechanical & Aerospace Engineering, Nanyang Technological University

Issued: 4th August 2026 **Submission deadline:** 15 August 2026, 23:59 SGT

Submit to: mark.wongkf@ntu.edu.sg with subject line Metasense-RE 4Q-response -
<Your Full Name>

1. Background

Thank you for your interest in the [Research Engineer position](#) on the Metasense programme. Metasense is a small research team under the guidance of Associate Professor Ng YK¹ and Dr. Mark Wong² seeking to translate the metasense technology into commercial applications. As this role has attracted a large number of strong applications and to ensure a fair and role-relevant selection process, shortlisted applicants are invited to complete a short technical response task.

Only candidates who submit appropriate solutions to this screening challenge will be invited to interview. Please read Section 2 before choosing your questions. The role's scope explains why these four problems were selected. This challenge is designed to:

- Identify candidates whose skills, career trajectory and working approach align with the intended scope of the role.
- Reveal how you work: we care more about your reasoning, engineering judgement, and honesty about your technical solutioning process than about a polished final result.
- Show you what the work actually looks like: each question is a simplified version of a problem on our current roadmap. If you join, you will work on the full version of at least one of them.

2. The role, and why these problems

Metasense is developing a clinical-grade, calibration-free cardiometabolic wearable. The scientific core is plethysmomorphometry: recovering hemodynamic biomarkers-blood pressure, arterial stiffness, pulse wave velocity, and vascular age - from the morphology of optical pulse waveforms and from pulse transit time, without a cuff and without per-patient calibration.

¹ <https://scholar.google.com.sg/citations?user=9QW1LYAAAAAJ&hl=en>

² <https://scholar.google.com.sg/citations?user=SVzzvSIAAAAAJ&hl=en>



The sensing front end applies a patented³ dual-point, multi-wavelength photoplethysmography (MW-2PPG) architecture: two spatially separated optical sites, each with several wavelengths, sampled simultaneously so that transit-time and morphology features can be extracted from the same heartbeat. The Research Engineer role sits at the intersection of three disciplines:

Domain	Description
Embedded / analog front end	Bring-up and characterization of optical analog front end architectures (AFEs) LED drivers and Transimpedance Amplifiers (TIA) gain-stage design and tuning; multi-slot acquisition scheduling; ambient and motion rejection; low-power firmware on Cortex-M class MCUs
Edge ML	Taking models that currently run on a workstation and making them run inside an MCU's memory and power budget - quantization, pruning, operator selection, and the accuracy trade-offs those force
Biosignal analysis	Beat segmentation, signal-quality assessment, morphological feature extraction, physics-constrained estimation of hemodynamic parameters, and honest validation against clinical reference standards

We do not expect one person to already be an expert in all three. We *are* looking for an engineer who is strong in at least one, and genuinely curious about the other two.

3. Challenge statement - what to do

Answer two of the four questions in Section 4. Choose the two that best represent how you would contribute to Metasense. Your choice is itself informative for us: pick the questions that showcase your strengths, not the ones that let you avoid difficulty. Every question can be answered using public data only. No synthetic data, proprietary datasets, or hardware purchases are required or expected, and none of the four questions requires physical hardware - simulation, calculation, or a written design are all acceptable. Partial results are welcome.

Submission format

Push your work to a GitHub repository and email us the link. Your repository must contain the working, runnable code. Provide the exact Python entry point (e.g., `python main.py` or `jupyter execute notebook.ipynb`) and include `requirements.txt`, `environment.yml`, or an equivalent. We will clone your repo on a clean machine and follow your instructions verbatim.

³ <https://patentscope.wipo.int/search/en/WO2024205510>



If the code does not run, we will not assess it. A README.md that, for each question you answered, states the approach you chose and why, the alternatives you considered and rejected. Your results, presented as numbers, tables, or plots. Reproducible outputs: Include a script, notebook, or make target that regenerates your headline results from scratch. Commit the output artefacts (plots, metric tables, logs) as well, so we can see your results even if our environment differs from yours.

On the use of AI coding assistants

Using LLM assistants is permitted under the following conditions:

- Disclose it in your README, for example: “Used Claude/Codex/Copilot for boilerplate and debugging,” or “Used it to draft the initial filter implementation, which I then rewrote.”
- Be able to defend every line with comments. If you are shortlisted, the interview will involve walking through your method, codes and explaining why specific choices were made. Vibed code you cannot explain will fail this screening.

Your work remains yours

This challenge is for applicant assessment only. We claim no rights over what you submit, we will not implement or incorporate it, and it will not be shared outside the selection panel. Keep your repository; it is a portfolio piece either way.

4. The questions

Question 1 - Embedded: low-power data storage for a wearable datalogger

The real problem. A continuous PPG/ECG datalogger can easily produce on the order of 10 Mb/minute of raw waveform data that must be written reliably to non-volatile storage - often without a DMA controller to help, because DMA channels are already committed to the acquisition front end, or because the target MCU class simply does not have one to spare.

Your task. Propose a non-DMA data-storage approach for a NAND/NOR flash or qSPI SD-card subsystem that sustains roughly 10 Mb/minute of continuous write throughput. Cover your buffering strategy (sizing, double-buffering, and how the CPU stays free for acquisition while writes are in flight), how you handle flash wear-leveling and write amplification, and how a dropped or delayed write would be detected and recovered without silently losing data. State your assumed MCU class and clock speed and show the throughput arithmetic behind your design - not just an assertion that it “should be fast enough.”



We are specifically looking for: realistic timing and buffer-sizing arithmetic, awareness of flash failure modes, and a genuine recovery strategy - on a clinical wearable, silent data loss is not a minor inconvenience.

Question 2 - Biosignal/Embedded: low-compute artefact rejection for battery-limited wearables

The real problem. Signal-quality assessment is usually built on models heavy enough to noticeably shorten battery life on a wearable, yet acquiring an artefact-corrupted signal is worse than acquiring none, since it silently corrupts every biomarker computed downstream. Reducing that compute cost is one of the more direct paths to a clinical-grade device people will actually wear all day.

Your task. Describe - and implement a concept in Python - an unsupervised, feature-based method for rejecting artefact-corrupted segments of a PPG or ECG signal, cheap enough to run continuously on a low-power microcontroller without a trained ML classifier in the loop. State which signals properties your method exploits and why, and estimate its compute cost (operations, memory) against a typical lightweight ML alternative.

We are specifically looking for: a method you can justify from first principles about the signal rather than a black box; realistic reasoning about what “cheap enough for a battery-powered wearable” means in practice; and honesty about the artefacts your approach would miss.

Question 3 – Bio signals/ML: a self-supervised, zero-shot signal-quality detector

The real problem. Hand-tuned signal-quality heuristics - thresholds on autocorrelation or autoregressive stability - are quick to run but work poorly on skin tones, motion artefacts, and sensor placements. Fully supervised quality classifiers need labelled data that usually does not exist at the point of deployment.

Your task. Sketch a Python design (pseudocode is fine; a working prototype is a bonus) for a method that segments a continuous PPG recording into beats and learns, in a zero-shot, self-supervised manner, to separate physiologically plausible, periodic segments from artefact-corrupted ones - without labelled training data. You are free to build this from classical signal statistics (autocorrelation, autoregression) or propose any self-supervised representation-learning ideas (SimSiam-style contrastive learning, adversarial

autoencoders etc.) – discuss and justify whichever direction you take. Sketch how you would validate the resulting quality labels against expert annotations without ever training them.

We are specifically looking for: a clear argument for why your proposed method should work, grounded in signal structure rather than architecture name-dropping, and a validation plan that could actually catch the method fooling itself.

Question 4 - Edge ML: fitting a model into a microcontroller's budget

The real problem. Models built and validated on a workstation eventually have to run inside a wearable MCU - inside its memory, flash, and power budget - and every simplification made to get there has an accuracy cost downstream.

Your task. Take any model that regresses a continuous physiological target (blood pressure or heart rate, say) from a windowed PPG or ECG segment, and compress it to plausibly fit a small Cortex-M-class budget: on the order of a few hundred KB of RAM and flash, integer-only inference. Apply at least one compression technique - quantization, pruning, or other methods - and report the accuracy cost against a held-out, subject-disjoint split. A random, per-window split leaks information and will read here as a red flag, not a shortcut.

We are specifically looking for: honest, quantified trade-offs rather than an assertion that “it still works,” and memory and latency accounting that is realistic rather than aspirational.

5. How we will assess your submission

Weighted approximately as follows:

Criterion	Weight	What we are reading for
Reasoning and justification	25%	Why this approach? What was rejected, and why? Is the thinking visible?
Correctness and rigor	25%	Does it work? Is the validation sound? Is the memory, power, or error accounting honest?
Code quality	25%	Readable, well-structured, and runnable by a stranger - not enterprise architecture, just clarity.
Communication and reporting	25%	Can we understand what you did and found from README alone - including the weaknesses stated plainly rather than left for us to discover?

We would rather see a negative result you understood than a positive result you cannot account for. If your model underperforms, or your control loop oscillates and you cannot fully fix it, write that up. It is not a penalty. Indefensible, unexplainable overclaims will fail this screening. Unsupported accuracy figures, benchmark comparisons against differently validated results, or a leaky split presented as a clean one will weigh more heavily against you than any technical shortfall.

6. Timeline and next steps

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Technical interview Will involve a walkthrough of your submission, plus discussion

If the problem statement is ambiguous, ask: mark.wongkf@ntu.edu.sg with subject line Metasense-RE 4Q-response - <Your Full Name> by 15 August. Asking a good clarifying question is a positive signal, not a negative one. Where genuine ambiguity exists and clarification will assist with your solutioning, we will be happy to provide so and establish some type of professional culture should you be offered the position. Where the statement is silent on some design decision, that is usually deliberate - make a reasonable choice, state it in your README, and move on.

Please note: Due to the volume of capable applicants, we will only progress applicants for interviews who submit by the deadline. If you do not submit or find the tasks unsuitable with your personal goals; we will not contact you further regarding this position. If circumstances make the deadline difficult - you are mid-thesis, you have a competing commitment, you are currently in notice period - write to us before 8th August.

We can usually accommodate a short extension where there is a reason, and we would rather do that than lose a good candidate to a calendar clash. We appreciate the time this asks of you, and we do not ask for it lightly. Whatever the outcome, you will leave this process with a piece of work worth keeping.

We look forward to reading your submissions. All the best!